

Refutation of conjecture 00000001109

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Verdict: FALSE

We refute conjecture 00000001109 as stated, under the universal reading that its own wording signals. For every *irreducible* finite non-crystallographic Coxeter group the degrees — hence the exponents — are pairwise distinct, so every exponent has multiplicity exactly 1 and the conjectured conclusion (“some exponent of multiplicity ≥ 2 ”) is false.

Scope (stated up front, and honestly). The refutation is *universal*: it holds for all irreducible finite non-crystallographic Coxeter groups, which is what the conjecture’s phrase “universality of multiplicity structure” (中文: 重数结构普遍性) says. It is *not* unconditional over arbitrary groups. Under an *existential* reading that also admits *reducible* non-crystallographic groups the conjecture is TRUE: the reducible group $A_1 \times H_3$ is non-crystallographic and has exponent multiset $\{1, 1, 5, 9\}$, so the exponent 1 has multiplicity 2. See Section 5.

1 The conjecture

Quoted verbatim from `conjectures/00000001109.md`:

English. Definition: The exponents of a Coxeter system (the characteristic polynomial). Conjecture: The exponent set of a non-crystallographic Coxeter group contains, besides $(h, 1)$, some exponent of multiplicity ≥ 2 (universality of multiplicity structure).

中文。 定义: Coxeter 系统的指数 (特征多项式)。猜想: 非晶体 Coxeter 群的指数集包含 $(h, 1)$ 之外的重数 ≥ 2 的指数 (重数结构普遍性)。

The claim says: for a non-crystallographic Coxeter group, apart from the special pair $(h, 1)$, there is an exponent whose multiplicity is at least 2. The parenthetical “universality of multiplicity structure” indicates that this is asserted as a general structural law across such groups, i.e. a *universal* statement over the class, not the mere existence of one example. We take the conjecture in that universal sense.

2 Conventions

Definition 1 (degrees, exponents, Coxeter number). For a finite Coxeter group W let $d_1 \leq d_2 \leq \dots \leq d_n$ be its *degrees* (the degrees of the fundamental polynomial invariants) and let h be its *Coxeter number*. The *exponents* of W are

$$e_i = d_i - 1 \quad (i = 1, \dots, n),$$

and h is the largest degree, $h = d_n$. The *multiplicity* of an exponent is its multiplicity as an element of the multiset $\{e_1, \dots, e_n\}$.

Since the exponents are the degrees minus one, distinctness of degrees is equivalent to distinctness of exponents. For a finite Coxeter group the exponents are exactly the roots of the characteristic polynomial; we use the equivalent degree description throughout.

Remark 2 ($(h, 1)$ is vacuous under the standard convention). Because $h = d_n$ is the largest *degree* and the exponents are the degrees minus one, the largest exponent is $h - 1 < h$. Hence h is *not* an exponent, and the phrase $(h, 1)$ in the conjecture does not correspond to an exponent at all: under the standard convention it is vacuous. Even on the generous reading in which one adjoins h to the exponent list, the conclusion still fails (Proposition 5 below).

3 The irreducible finite non-crystallographic types

The complete list of irreducible finite Coxeter types that are non-crystallographic is

$$H_3, \quad H_4, \quad I_2(m) \quad (m \geq 5).$$

Here $I_2(3) = A_2$, $I_2(4) = B_2$, $I_2(6) = G_2$ are crystallographic, and $I_2(2) = A_1 \times A_1$ is reducible; hence the irreducible *non-crystallographic* I_2 cases are exactly $m \geq 5$. (This classification, and the exponent tables below, are classical; see Humphreys [1] and Bourbaki [2].)

Type	Degrees	h	Exponents	Multiplicities
H_3	2, 6, 10	10	1, 5, 9	all 1
H_4	2, 12, 20, 30	30	1, 11, 19, 29	all 1
$I_2(m), m \geq 5$	2, m	m	1, $m - 1$	all 1

In the last row the two exponents 1 and $m - 1$ coincide exactly when $m = 2$; for $m \geq 5$ we have $m - 1 \geq 4$, so $1 \neq m - 1$. In every row the exponents are pairwise distinct, so each occurs with multiplicity 1.

4 The argument

Theorem 3. *Let W be an irreducible finite non-crystallographic Coxeter group. Then no exponent of W has multiplicity ≥ 2 .*

证明. By the classification of irreducible finite Coxeter types, W is one of H_3 , H_4 , or $I_2(m)$ with $m \geq 5$. From the exponent tables of Section 3:

- H_3 : exponents 1, 5, 9, which are pairwise distinct;
- H_4 : exponents 1, 11, 19, 29, pairwise distinct;
- $I_2(m)$, $m \geq 5$: exponents 1 and $m - 1$, distinct because $m - 1 \geq 4 > 1$.

In each case the exponents form a set with no repetitions, i.e. a multiset of multiplicity 1 everywhere. Therefore no exponent has multiplicity ≥ 2 . $\square$

Corollary 4. *Conjecture 00000001109, read universally over irreducible finite non-crystallographic Coxeter groups, is false.*

证明. Take $W = H_3$. Its exponent set is $\{1, 5, 9\}$; since $h = 10$, the pair $(h, 1)$ contributes nothing, and the remaining exponents 5, 9 each have multiplicity 1. So the exponent set contains no exponent of multiplicity ≥ 2 , contradicting the conjecture's conclusion. The same contradiction is obtained from H_4 (exponents 1, 11, 19, 29) and from every $I_2(m)$ with $m \geq 5$ (exponents 1, $m - 1$). $\square$

Proposition 5. *Adjoining h to the exponent list does not create a repeated exponent:*

$$H_3 : \{1, 5, 9, 10\}, \quad H_4 : \{1, 11, 19, 29, 30\}$$

are both all-distinct, and for $I_2(m)$ the list $\{1, m - 1, m\}$ is all-distinct for $m \geq 5$.

证明. Immediate from the tables: 10 differs from 1, 5, 9; 30 differs from 1, 11, 19, 29; and for $m \geq 5$ the three numbers 1, $m - 1 \geq 4$, and m are pairwise distinct (indeed $m - 1 \neq m$). $\square$

5 Honest scope: universal versus existential reading

The refutation above is a *universal* statement over the irreducible finite non-crystallographic types. We state clearly where the boundary of the claim lies.

Theorem 6 (Reducibility caveat). *Under an existential reading that admits reducible non-crystallographic Coxeter groups, conjecture 00000001109 is TRUE. Explicitly, the reducible group $A_1 \times H_3$ is non-crystallographic and has exponent multiset*

$$\{1\} \sqcup \{1, 5, 9\} = \{1, 1, 5, 9\},$$

so the exponent 1 has multiplicity $2 \geq 2$.

证明. Exponents add over direct products of Coxeter groups: the exponents of $W_1 \times W_2$ are the exponents of W_1 together with those of W_2 . The group A_1 has a single exponent, 1 (degree 2), and H_3 has exponents 1, 5, 9. Hence $A_1 \times H_3$ has exponent multiset $\{1, 1, 5, 9\}$, in which 1 occurs twice. The direct product is non-crystallographic because H_3 is, and it is not irreducible. Thus an existential reading of the conjecture — “there exists a non-crystallographic Coxeter group with a repeated exponent” — is satisfied. $\square$

Remark 7 (Why the universal reading is the intended one). The conjecture does not say “there exists a non-crystallographic Coxeter group with ...”; it says the exponent set *contains* such an exponent and labels this “universality of multiplicity structure” (重数结构普遍性). “Universality” is a statement about the class of objects, so the natural reading is universal over non-crystallographic Coxeter groups. Under the universal reading the conjecture is false, by Corollary 4; the reducible counterexample (Theorem 6) shows that the refutation depends on this reading and would not survive an existential reformulation that admits reducible groups. We prefer to state this openly rather than leave it implicit.

Remark 8 (On $(h, 1)$). As noted in Remark 2, under the standard convention h is a degree and not an exponent, so the phrase $(h, 1)$ is vacuous. If one nevertheless adjoins h to the exponent list, the conclusion still fails (Proposition 5). So the refutation does not hinge on the interpretation of $(h, 1)$: it fails under either reading, while the reducibility issue above is the genuine scope boundary.

6 Reproducibility

Python. The script `reproduce.py` (Python 3 standard library only) encodes the exponent lists of H_3 , H_4 , and $I_2(m)$ for $m = 5, \dots, 12$, computes the multiplicities, asserts that every exponent has multiplicity 1, and asserts that the reducible case $A_1 \times H_3$ has a repeated exponent (so the scope boundary is visible in code). It prints `PASS` and exits 0 exactly when every check holds:

```
$ python3 reproduce.py
... (tables, multiplicity checks, scope check) ...
PASS: all checks verified; conjecture 00000001109 is FALSE ...
```

Lean 4. The accompanying project `lean4/` formalises the finite multiplicity arithmetic in core Lean only (`import Std`, no `Mathlib`, no `sorry`, no `native_decide`). It proves, for the concrete lists,

$$\neg \exists x, x \in \{1, 5, 9\} \wedge 2 \leq \#\{1, 5, 9\}(x)$$

and the analogous statement for $\{1, 11, 19, 29\}$ (axiom-free, by `decide`), the general lemma that a duplicate-free list has every member with count 1, the parametric statement for $I_2(m)$, and — openly — the reducibility boundary $\text{count}_1(\{1, 1, 5, 9\}) = 2$. It builds with `lake build` and is audited by `lake env lean Check.lean`, which reports no `sorryAx`. See `lean4/README.md` for the statement table and the scope section.

7 Cited inputs, not reproved

The classification of irreducible finite Coxeter groups, the fact that the exponents are the degrees minus one, and the exponent tables for H_3 , H_4 , and $I_2(m)$ are classical. They are cited here, not reproved:

- J. E. Humphreys, *Reflection Groups and Coxeter Groups*, Cambridge Studies in Advanced Mathematics 29, Cambridge University Press, 1990 (degrees, exponents, the classification).
- N. Bourbaki, *Groupes et algèbres de Lie*, Chapitres IV–VI, Hermann, Paris, 1968 (Coxeter groups and their invariant theory).

The present submission contributes the observation that the multiplicity structure is trivial (all multiplicities 1) for every irreducible finite non-crystallographic type, hence the conjectured “universality of multiplicity structure” fails; and it records explicitly the reducible exception that bounds the claim.

参考文献

- [1] J. E. Humphreys, *Reflection Groups and Coxeter Groups*, Cambridge Studies in Advanced Mathematics 29, Cambridge University Press, 1990.
- [2] N. Bourbaki, *Groupes et algèbres de Lie*, Chapitres IV–VI, Hermann, Paris, 1968.